

Applied Math 50: Introduction to Applied Mathematics

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Overview of today

- 1. What is applied mathematics and why should you take this course?**
- 2. Course organization**
- 3. Topic for the first week: a simple mathematical model for sport competitions**

History of AM50

- **Started in 2009**
- **Most applied math courses focus primarily on techniques and methods (Math 1b, Applied Math 21a, *etc.*)**
- **This course has been designed to provide a broader overview of what it means to study applied mathematics, by showing through examples about how simple mathematical ideas have had a large impact**

Isn't applied math ...

... kind of like engineering?

Both seem interested in solving practical problems.

Both use math and computer simulation.

Isn't applied math ...

... kind of like engineering?

... kind of like physics?

Both are interested in understanding the physical world.

You can take courses in mechanics (*e.g.* Newton's laws) in physics and math departments.

Isn't applied math ...

... kind of like engineering?

... kind of like physics?

... a subset of math?

Why make a distinction at all?

Many universities just have math departments.

Quotes

“I describe applied mathematics as the bridge connecting pure mathematics with science and technology.”

W. Prager, 1972

“One should not judge applied mathematicians by their contributions to mathematics any more than one judges historians by their contributions to history. Much creative effort goes into interpretation. To try to use mathematical ideas on problems which at first sight seem too complex or obscurely stated to fit any pattern is, for many people, an exciting challenge and it is this which I feel to be the essence of applied mathematics.”

R. Brockett, 1972

“The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift for which we neither understand nor deserve. We should be grateful for it, and hope that it will remain valid ... to wide branches of learning.”

E. Wigner, 1960

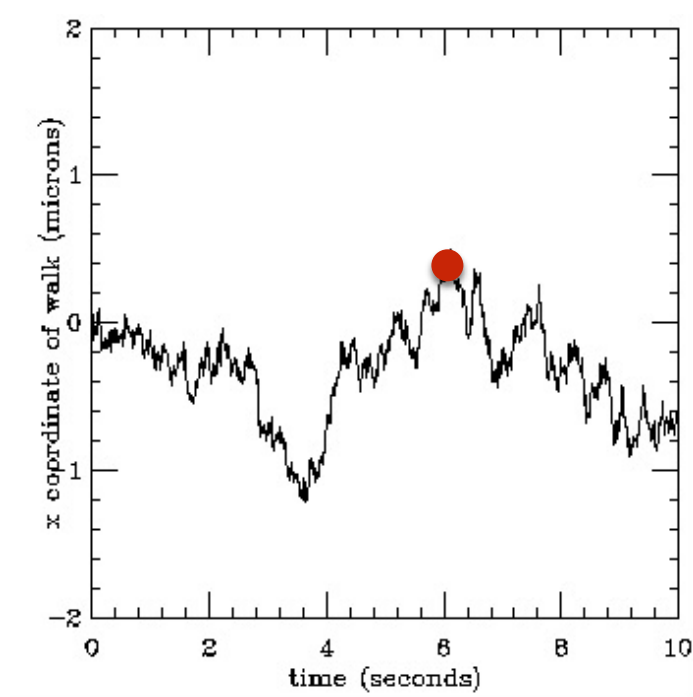
Summary

- **Applied Mathematics is an inherently interdisciplinary subject, which covers a wide spectrum of scientific activities. It is the mathematics dealing with problems arising in the physical, life and social sciences as well as in engineering, and provides a broad qualitative and quantitative background for use in these fields.**
- **The methods of mathematical modeling and analysis provide a unification and mutual enrichment of ideas from many different areas, and a deeper understanding of the fields to which it is applied. Applied Mathematics draws upon concepts and methods of mathematics from the fields of application and in turn, brings ideas, techniques and scientific knowledge back to influence the development of mathematics.**

Common patterns, common processes



Ten year plot
of Dow
Jones
Industrial
Average



Random
motion of a
diffusing
particle

The Problem of the Random Walk.

CAN any of your readers refer me to a work wherein I should find a solution of the following problem, or failing the knowledge of any existing solution provide me with an original one? I should be extremely grateful for aid in the matter.

A man starts from a point O and walks l yards in a straight line; he then turns through any angle whatever and walks another l yards in a second straight line. He repeats this process n times. I require the probability that after these n stretches he is at a distance between r and $r + \delta r$ from his starting point, O.

The problem is one of considerable interest, but I have only succeeded in obtaining an integrated solution for *two* stretches. I think, however, that a solution ought to be found, if only in the form of a series in powers of $1/n$, when n is large.

KARL PEARSON.

The Gables, East Ilsley, Berks.

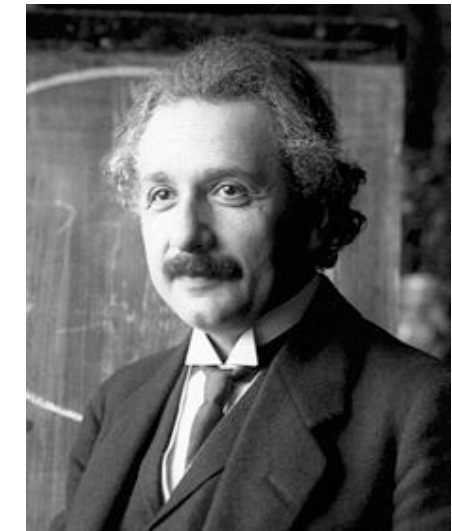


Common patterns, common processes

ON THE MOVEMENT OF SMALL PARTICLES SUSPENDED IN STATIONARY LIQUIDS REQUIRED BY THE MOLECULAR-KINETIC THEORY OF HEAT

by A. Einstein

[*Annalen der Physik* 17 (1905): 549-560]



THÉORIE
DE
LA SPÉCULATION,

PAR M. L. BACHELIER.

INTRODUCTION.

Les influences qui déterminent les mouvements de la Bourse sont innombrables, des événements passés, actuels ou même escomptables, ne présentant souvent aucun rapport apparent avec ses variations, se répercutent sur son cours.

A côté des causes en quelque sorte naturelles des variations, interviennent aussi des causes factices : la Bourse agit sur elle-même et le mouvement actuel est fonction, non seulement des mouvements antérieurs, mais aussi de la position de place.

La détermination de ces mouvements se subordonne à un nombre infini de facteurs : il est dès lors impossible d'en espérer la prévision mathématique. Les opinions contradictoires relatives à ces variations se partagent si bien qu'au même instant les acheteurs croient à la hausse et les vendeurs à la baisse.

Le Calcul des probabilités ne pourra sans doute jamais s'appliquer aux mouvements de la cote et la dynamique de la Bourse ne sera jamais une science exacte.

Mais il est possible d'étudier mathématiquement l'état statique du marché à un instant donné, c'est-à-dire d'établir la loi de probabilité des variations de cours qu'admet à cet instant le marché. Si le marché, en effet, ne prévoit pas les mouvements, il les considère comme étant



L. BACHELIER

Théorie de la spéculation

Annales scientifiques de l'É.N.S. 3^e série, tome 17 (1900), p. 21-86.

http://www.numdam.org/item?id=ASENS_1900_3_17_21_0

Common patterns, common processes

L'expression définitive de la probabilité est donc

$$p = \frac{1}{\sqrt{2\pi}k\sqrt{t}} e^{-\frac{x^2}{2k^2t}}$$

L'espérance mathématique

$$\int_0^\infty px \, dx = k\sqrt{t}$$

est proportionnelle à la racine carrée du temps.

**Bachelier's solution
motivated by stock
price variations**

$$f(x, t) = \frac{n}{\sqrt{4\pi D}} \frac{e^{-\frac{x^2}{4Dt}}}{\sqrt{t}}$$

$$\lambda_x = \sqrt{2Dt}.$$

**Einstein's solution
motivated by random
particle motion**

Common patterns, common processes

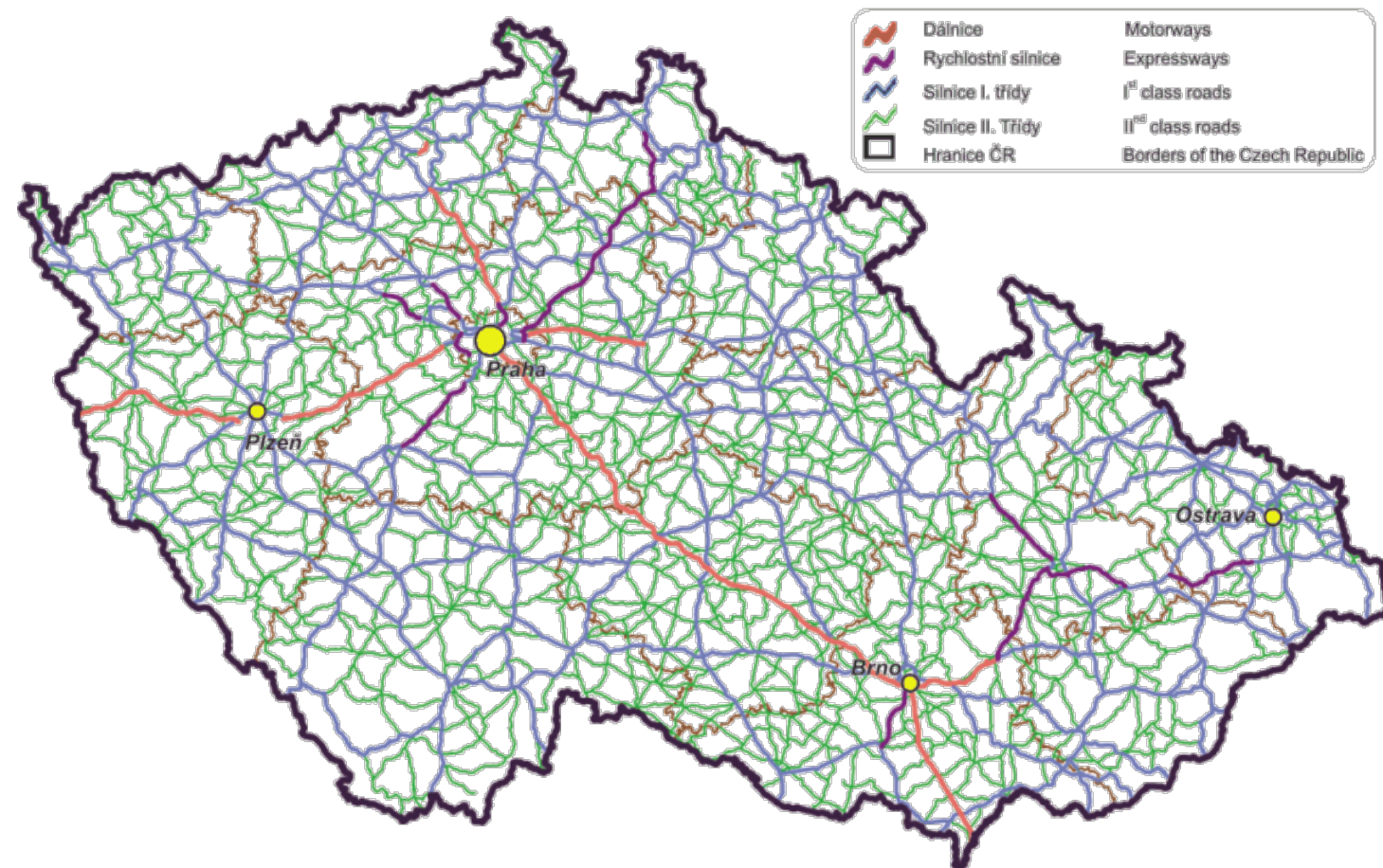


**Drainage network in
Baja California salt flat**

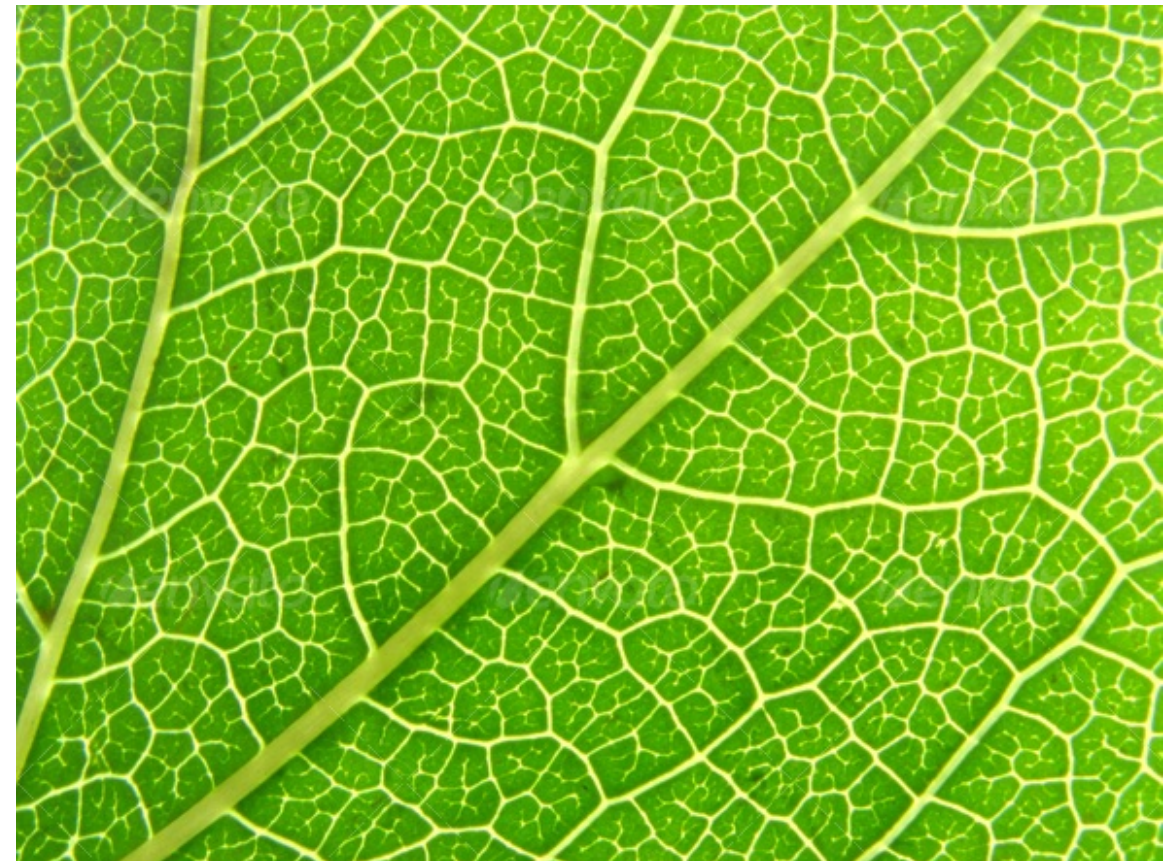


Tree branches

Common patterns, common processes

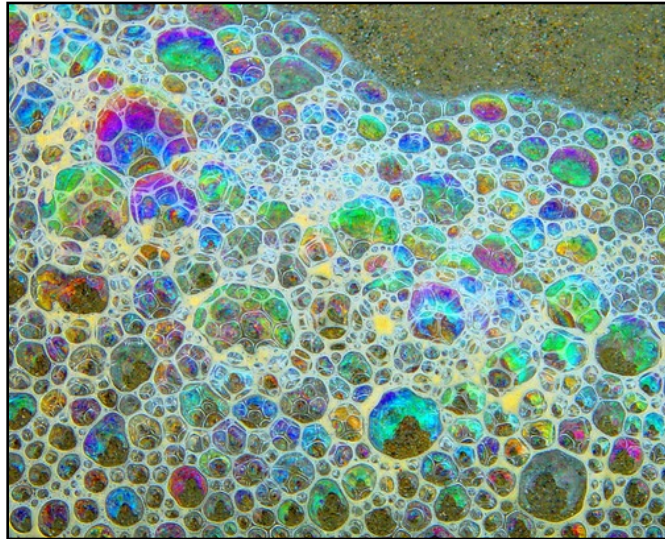


Highways in Czech Republic



Leaf veins

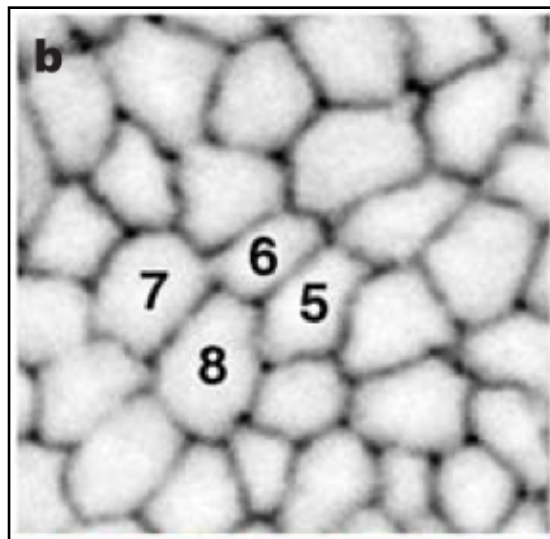
Common patterns, common processes



Soap foam



Stone polygons
common in regions of permafrost

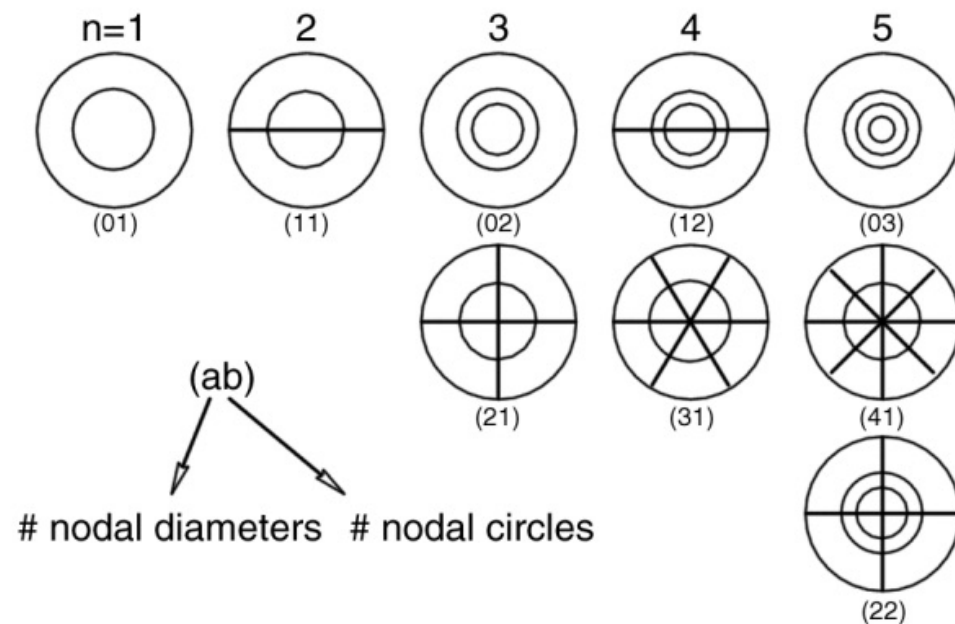


**Cell boundaries in
fruit fly wing**

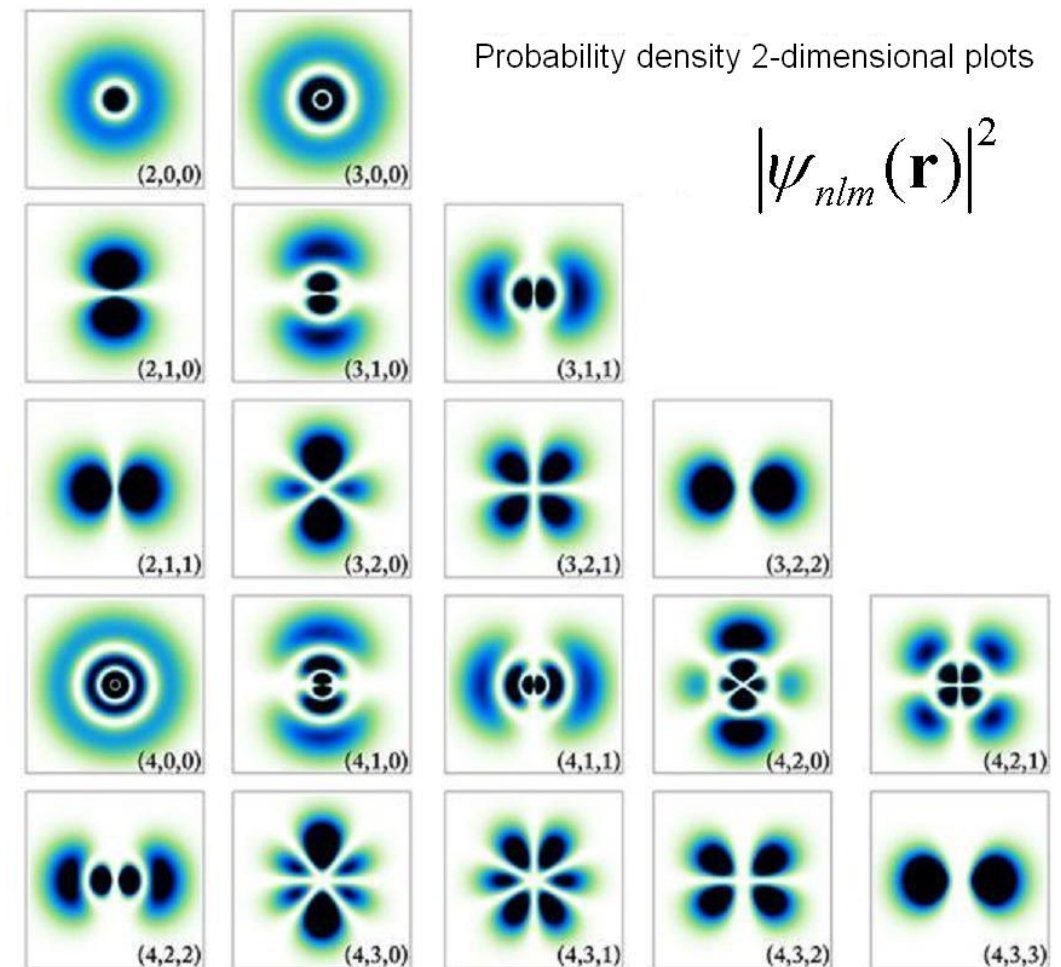


Polygonal basalt columns
*Giants Causeway, Northern
Island*

Common patterns, common processes



Oscillation modes of a drum surface



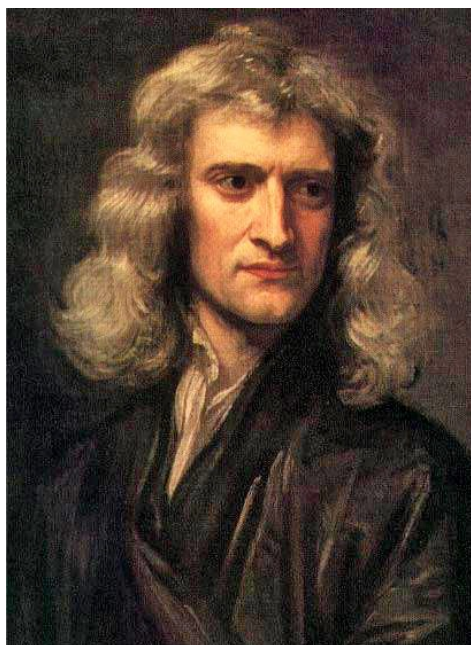
Electron orbitals of a hydrogen atom

Applied mathematics: a centuries-old approach



Lord Rayleigh (1842–1919)

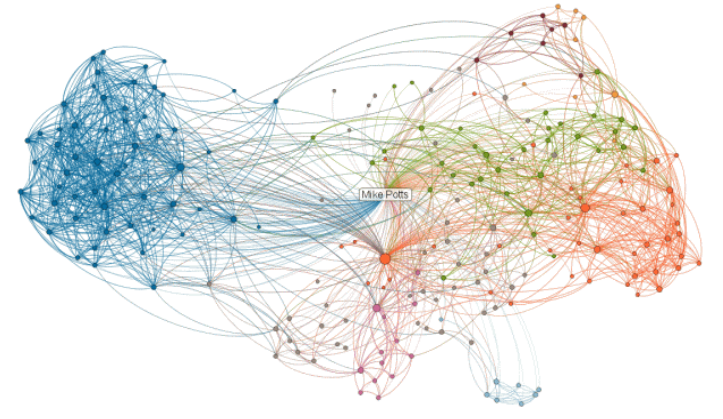
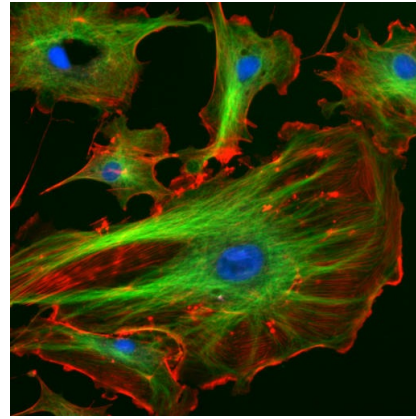
Professor of Natural Philosophy at the University of Cambridge (1874–1884). Made major mathematical contributions to fluid dynamics, physics, and acoustics.



Isaac Newton (1642–1727)

Laid the mathematical foundations of classical mechanics through *Philosophiæ Naturalis Principia Mathematica* published in 1687

21st century applied math



- **21st century science and technology: immense and highly specialized**
- **Mathematics underpins almost every field; there is a great need for people who can apply mathematics to important practical issues**
- **Many applied mathematicians have careers spanning multiple disparate fields**
- **An excellent discipline if you're interested in everything, and want to understand the deep connections between fields**

2. Course organization

AM50: Course objectives

- **Expose you to a wide variety of topics where applied math is used**
- **Through reading great mathematical papers, show how simple mathematical models can have large impact on different fields of human inquiry**
- **Learn practical skills:**
 - **Reading for main points in math papers**
 - **Programming in Matlab**
 - **Communicating mathematical ideas through oral and written presentations**
 - **Working in a group**

Prerequisites

- **Syllabus lists AM21a (possibly taken concurrently) as a prerequisite**
- **Same prerequisite as previous years**
- **Some ideas from AM21a but not many; in previous years students doing 1b have been fine**
- **Course mainly aimed at freshmen and sophomores**
- **Programming experience is NOT a prerequisite**

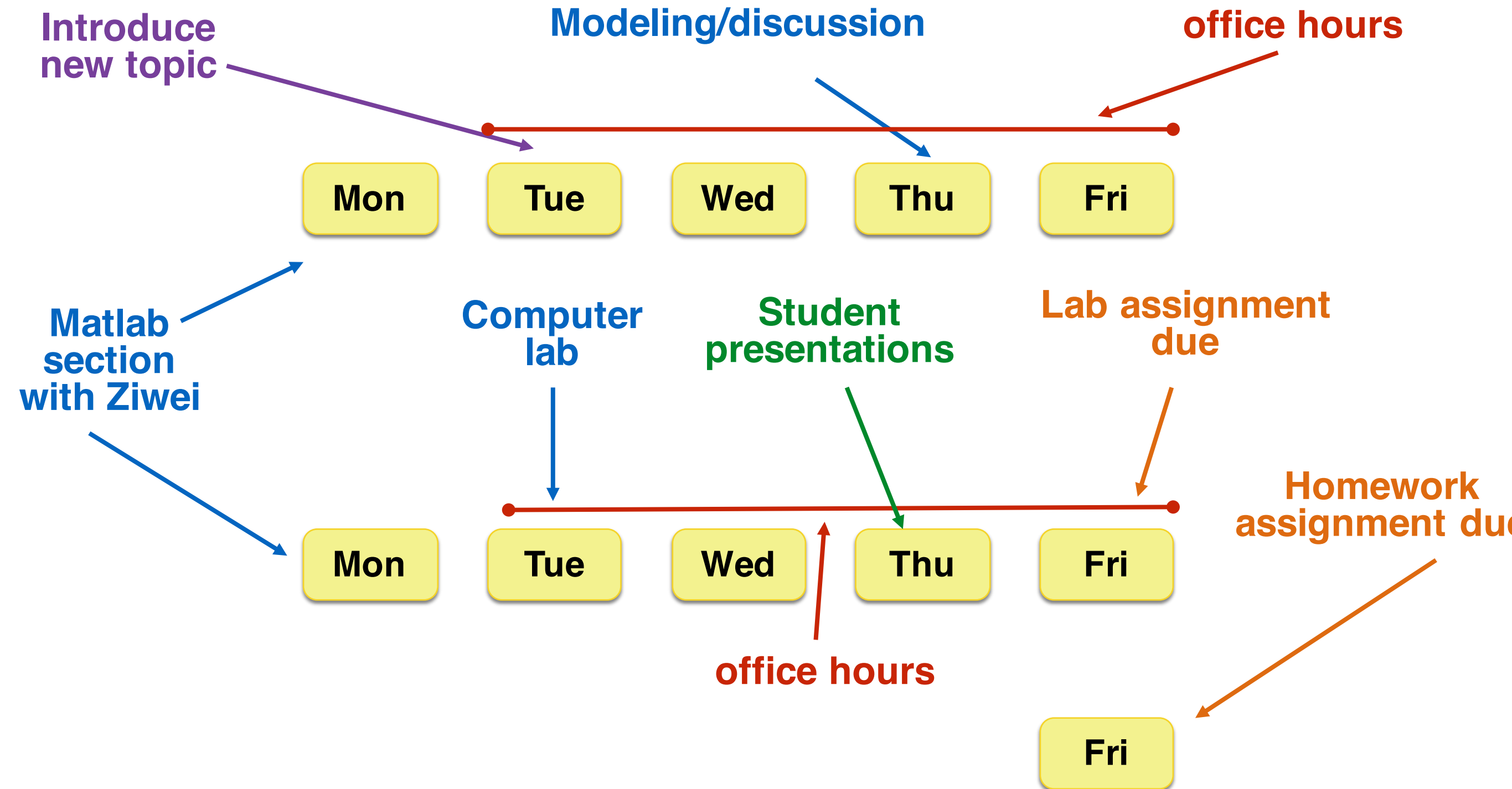
Course layout

Jan 24	Introduction	Mar 14	Spring Break
Jan 31	Topic 1: Epidemiology	Mar 21	Topic 4: Information theory
Feb 7		Mar 28	
Feb 14	Topic 2: Imaging	Apr 4	Topic 5: Game theory
Feb 21		Apr 11	
Feb 28	Topic 3: Random walks	Apr 18	Topic 6: Chaos or Projects
Mar 7		Apr 25	

Each topic will include:

- **Class 1: introductory class where a new topic will be presented via a discussion of a math paper.**
- **Class 2: the topic will be explored in more detail. This class will usually include a short modeling activity, which will be done in small groups.**
- **Class 3: a lab where ideas from the paper are simulated in Matlab.**
- **Class 4: student presentations of papers, peer critiques of presentations**

General two-week topic schedule



Introductory week

Today's
intro
lecture

Modeling/discussion

HW 0, Lab 0
posted

Mon

Tue

Wed

Thu

Fri

Lab 0 with Ziwei

Mon

Fri

HW 0, Lab 0 due

Matlab bootcamp

<https://wiki.harvard.edu/confluence/display/fasmatlab/MATLAB+Boot+Camp+Home>

Homeworks/labs: some programming in Matlab

- **Computer programming is a great way to develop and test mathematical models**
- **We will make use of Matlab in this course**
- **We strongly encourage absolute beginners to take the course, and we will help as much as possible**
- **Why Matlab? Used by many courses at Harvard, used by applied mathematicians and scientists, well-supported at Harvard**

Note: for AM concentrators, this course counts toward the modeling requirement, not the computing requirement

Collaboration policy

- **Collaboration on the homework and labs is encouraged, but each student must write solutions *in their own words***
- **If you end up copying verbatim large sections of text, equations, or code from a friend, that's not okay**
- **Full academic integrity policy on website.**

Writing Mathematics

4.7 Denseness of $\mathbb{Q}$.

If $a, b \in \mathbb{R}$ and $a < b$, then there is a rational $r \in \mathbb{Q}$ such that $a < r < b$.

Proof

We need to show that $a < \frac{m}{n} < b$ for some integers m and n where $n > 0$, and thus we need

$$an < m < bn. \quad (1)$$

Since $b - a > 0$, the Archimedean property shows that there exists an $n \in \mathbb{N}$ such that $n(b - a) > 1$. Since $bn - an > 1$, it is fairly evident that there is an integer m between an and bn , so that (1) holds. However, the proof that such an m exists is a little delicate. We argue as follows. By the Archimedean property again, there exists an integer $k > \max\{|an|, |bn|\}$, so that

$$-k < an < bn < k.$$

Then the set $\{j \in \mathbb{Z} : -k < j \leq k \text{ and } an < j\}$ is finite and nonempty and we can set

$$m = \min\{j \in \mathbb{Z} : -k < j \leq k \text{ and } an < j\}.$$

Sample from *Introduction to Analysis* by K. Ross

- If you look at any good science textbook, you'll see there's a standard way of writing mathematical equations
- Equations should form part of full sentences, and should go with the flow of text
- We strongly encourage you to practice this in AM50

Writing Mathematics

- **LaTeX is a free software package that is widely used for typesetting high-quality scientific documents**
- **It is exceptionally good for typesetting mathematical equations**
- **We encourage you to try it**
- **Links will be posted on the website**

```
\documentclass[12pt]{article}

\begin{document}
The geometric series summation formula is
\begin{equation}
\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}
\end{equation}
where  $a$  is a real number satisfying  $|a| < 1$ .
\end{document}
```

Input file of LaTeX commands

The geometric series summation formula is

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a} \quad (1)$$

where a is a real number satisfying $|a| < 1$.

PDF output file

Course discussion on Piazza

- We will use Piazza for online discussion
- Sign-up link will be posted on website
- Enrolled students will receive a sign-up email

The screenshot shows the Piazza Q&A interface for the AM 205 course. The top navigation bar includes links for 'Q & A', 'Resources', 'Statistics', and 'Manage Class'. The left sidebar lists various posts, including 'BFGS: lecture note vs wikipedia', 'Office hours about projects', 'Prob2c', 'Problem 2 Potential Energy', 'Prob3 Putting Constraint on Minimiser', 'Poster printing for CS poster session', 'Poster submission', 'Check Derivatives for Problem 2', 'Q2: Multiple Solutions', 'Question about scipy optimize', 'Q1: (c) BFGS tolerance needed lowerin...', 'Prob4', and 'Problem 2 - Least Action'. The main content area displays a question titled 'Prob2c' asking about the difference between BFGS and L-BFGS-B methods. Below the question, there is a student answer endorsed by the instructor (Chris Rycroft) and an instructor answer explaining the BFGS method. The bottom section shows follow-up discussions and a summary of the thread's activity, including the average response time (26 min) and the number of answers (1).

Paper presentations

- **Twice during the semester, each student will give roughly ten-minute presentation about a mathematical paper.**
- **The presentations can be done in pairs**
- **Your classmates will give you verbal/written feedback, which you can reflect on as you work on your second presentation.**
- **We have a database of papers related to each topic that you can choose from, or you can find a paper yourself**
- **We'll announce exact length/format once we have accurate enrollment numbers**

Final project

- **The final project will involve developing and testing a mathematical model**
- **Students can work in pairs**
- **We'll start discussing project ideas before Spring Break**
- **Will be due at the end of the semester**

Grading

15% : Participation

Engagement in class activities, student critiques of presentations, contributing to learning community

15% : Paper presentations in class

40% : Homework / computer lab assignments

Can drop one homework OR one lab

30% : Final modeling project

3. A simple mathematical model for sport competitions

2004 World Series: Go SOX!



Isn't there a lot of luck in this game?
Did the better team really win?

Mosteller's model



Charles Frederick Mosteller (1916–2006)

Famous 20th century statistician, chair of the Harvard Statistics department (1957–1971), and avid Red Sox fan

JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION

Number 259

SEPTEMBER 1952

Volume 47

THE WORLD SERIES COMPETITION*

FREDERICK MOSTELLER
Harvard University

Wrote paper in 1952 analyzing the baseball world series (available on course website)

Basic idea

Many sporting events have “play-offs”, designed to find out who is the better team:

- **Football (Superbowl) — play one game**
- **Baseball (World Series) — best out of 7 games**
- **Tennis — 3 or 5 sets in a match**
- **Snooker (World Championship) — 35 frames in final**



Mosteller's question

IF WE compare pairs of teams, products, drugs, or persons on the basis of a fixed number of binomial trials, and identify the member of the pair that wins the majority of trials as the better, we may be in error. For example, if the members of a pair are evenly matched, a decision on the basis of performance is equivalent to coin-flipping. On the other hand, if one of a pair is actually better, it is more likely that the better member will also be the winner. If we carry out such comparisons on many pairs under roughly comparable conditions, it is of interest to estimate the over-all effectiveness of the decision technique in the kinds of situations that have occurred in practice. Data from the World Series are available to illustrate many facets of this type of problem.

About World Series time each year most fans are wondering which team will win. The author is no exception, but he has also wondered about another question: Will the Series be very effective in identifying the better team?

Simple model for the World Series

As a simple model for the League play-offs, we might suppose that one team has probability p of winning single games, and that the other has probability $1 - p$ and further, that p remains the same for all games in the series under consideration. The reader may easily think of arguments against the assumption of the constancy of p from game to game (pitchers and ball parks are examples of variables that could contribute to variation in p). We will provide some evidence on this issue later.

- **Assume probability for a given team to win is p**
- **Assume probability is constant throughout the series**

Simple model

Over a series, playing baseball is like flipping an unfair coin



Is that
reasonable
?

Mosteller's model

- **Assume outcome of each game is the same as flipping an unfair coin**
- **The better team is the one with the higher probability of winning a single coin flip**
- **The point of playing a “series” is that you would like to amplify the chance the better team wins**
- **Intuitively, it seems clear that the more games you play, the more it will amplify the advantage of the better team**
- **But how good is it? What is the chance a better team will win a 1/3/5/7 game series?**

Binomial trials

- Suppose you are given a coin, and the probability of getting a head is p . What is the probability of getting m heads out of N trials?

Binomial trials

- Consider $N = 3$
- 8 possible ordered outcomes of 3 coin flips:

HHH THH
HHT THT
HTH TTH
HTT TTT

0 heads	1 head	2 heads	3 heads
1	3	3	1

- Probability of two heads and then one tail:
$$p \cdot p \cdot (1 - p) = p^2(1 - p)$$
- Probability of getting two heads in any order:
$$3 p^2(1 - p)$$

Binomial distribution

$$P(N, m) = \frac{N!}{(N - m)! m!} p^m (1 - p)^{N - m}$$

For example, for $N = 3$ and $m = 2$,

$$\begin{aligned} P(3, 2) &= \frac{3!}{(3 - 2)! 2!} p^2 (1 - p)^{3 - 2} \\ &= 3 p^2 (1 - p) \end{aligned}$$

Binomial trials

- Suppose you are given a coin, and the probability of getting a head is p . In N trials the probability of getting exactly m heads is

$$P(N, m) = \frac{N!}{(N - m)! m!} p^m (1 - p)^{N - m}$$

- Alternative notation

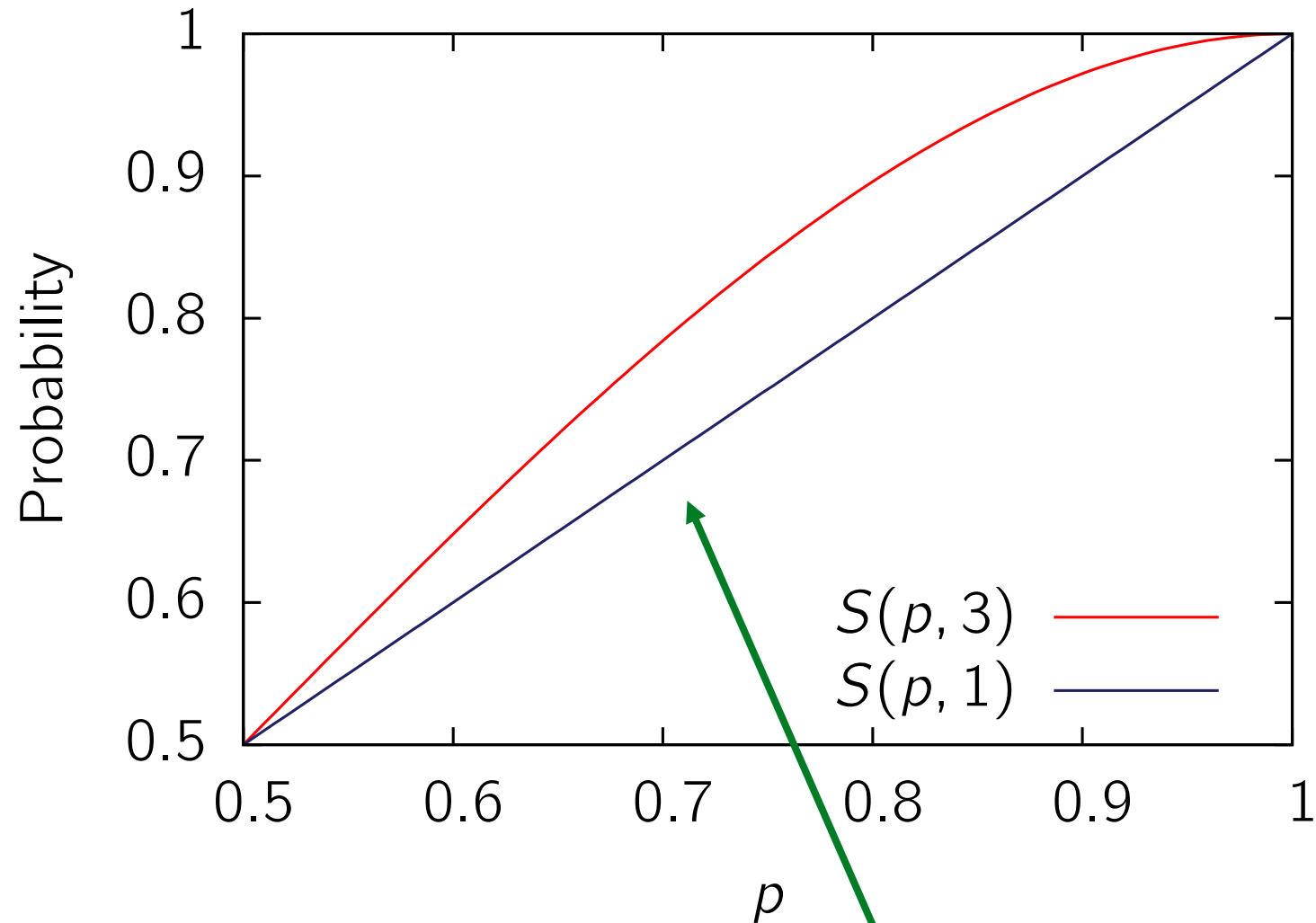
$$P(N, m) = \binom{N}{m} p^m (1 - p)^{N - m}$$

“ N choose m ”: the number of
ways to choose m items from N
things



Success probability

- Define $S(p, N)$ to be the probability of winning an N game series



- Then

$$\begin{aligned} S(p, 3) &= \Pr(\text{Win } 3) + \Pr(\text{Win } 2) \\ &= P(3,3) + P(3,2) = p^3 + 3p^2(1 - p) \end{aligned}$$

No series

- Winning combinations:

Mosteller's notation

B: better team wins

W: worse team wins

BBB, **BBW**, **BWB**, **WBB**